

SINGAPORE SELETAR, Singapore

WMO: 486990

Lat: 1.417N

Lon: 103.868E

Elev: 36

StdP: 14.68

Time Zone: 8.00 (E08)

Period: 04-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	74.1	74.9	66.2	96.7	88.7	67.8	102.3	87.2	20.4	84.5	18.7	84.5	5.3	220	0.373

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6	11.2	93.5	77.2	93.0	77.2	91.6	77.3	80.5	86.3	80.1	85.8	79.8	85.6	8.2	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
79.1	151.4	82.6	78.9	150.2	82.5	78.5	148.3	82.4	44.3	87.4	43.8	86.6	43.4	85.8	82.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 16.6	14.8	13.3	72.8	95.8	0.9	2.2	72.2	97.5	71.7	98.8	71.2	100.0	70.5	101.7		
(5)			69.7	81.8	0.9	0.6	69.0	82.3	68.5	82.6	68.0	83.0	67.4	83.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	83.5	81.9	83.0	83.8	84.3	84.6	84.3	83.9	83.8	83.6	83.7	82.8	82.1	
	DBStd	2.25	2.17	1.76	2.07	1.96	1.96	2.43	2.39	2.37	2.13	1.99	1.74	2.06	
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0	
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0	
	CDD50	12221	989	924	1046	1029	1073	1030	1050	1047	1009	1044	984	996	
	CDD65	6746	524	504	581	579	608	580	585	582	559	579	534	531	
	CDH74	73472	5004	5079	6160	6399	7126	6903	6830	6737	6331	6379	5376	5148	
	CDH80	26960	1478	1716	2229	2455	2910	2837	2711	2629	2441	2383	1695	1476	
Wind	WSAvg	5.7	8.2	9.0	6.6	4.5	4.2	4.5	5.4	5.6	4.9	4.4	4.4	6.3	
Precipitation	PrecAvg	88.1	7.2	6.8	7.8	7.6	7.2	6.7	5.9	4.5	6.9	6.9	9.1	11.4	
	PrecMax	116.9	18.0	16.6	15.3	13.8	13.4	10.8	9.7	7.9	20.2	14.5	16.3	21.6	
	PrecMin	66.4	1.0	0.1	1.7	1.5	3.6	2.9	0.8	1.3	2.4	1.8	2.8	3.1	
	PrecStd	15.2	4.2	4.1	4.0	3.2	2.8	2.2	2.6	2.0	4.4	3.6	4.0	5.2	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	91.7	93.3	93.5	94.8	94.6	94.8	93.4	93.4	93.3	93.6	91.8	91.8	
		MCWB	76.2	76.3	76.7	77.9	78.6	76.8	77.4	76.5	76.7	76.3	77.5	77.0	
	2%	DB	89.8	91.4	91.8	93.0	92.8	93.0	91.6	91.6	91.5	91.7	91.0	89.8	
		MCWB	76.5	76.2	76.6	77.8	78.3	77.3	77.4	77.0	76.8	76.5	77.4	77.0	
	5%	DB	88.1	89.7	91.1	91.4	91.3	91.4	90.0	90.0	89.9	91.0	89.5	88.1	
		MCWB	76.7	76.1	76.6	77.9	78.3	77.7	77.2	76.9	76.8	76.6	77.3	77.3	
	10%	DB	87.4	88.1	89.5	89.7	89.7	89.7	89.4	89.3	89.3	89.4	87.8	86.4	
		MCWB	76.6	76.1	76.7	77.8	78.2	77.7	77.2	76.9	76.9	77.0	77.8	77.2	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	79.9	79.2	80.0	80.8	81.3	80.9	80.1	80.0	80.1	80.4	80.1	80.1
			MCDB	84.9	85.5	86.9	86.8	87.3	86.4	85.6	85.7	85.6	85.3	85.2	84.9
2%		WB	79.0	78.3	79.2	80.0	80.5	80.1	79.7	79.3	79.4	79.7	79.6	79.2	
		MCDB	83.7	84.8	85.5	86.1	86.6	85.9	85.4	85.3	85.1	84.9	84.7	84.0	
5%		WB	78.3	77.8	78.7	79.6	80.0	79.8	79.1	78.9	78.9	79.1	79.1	78.8	
		MCDB	83.3	84.4	84.7	85.7	86.0	85.6	85.1	85.0	84.7	84.6	84.1	83.5	
10%		WB	77.7	77.2	78.1	79.1	79.7	79.2	78.7	78.3	78.3	78.6	78.6	78.3	
		MCDB	83.1	83.8	84.2	85.1	85.6	85.2	84.9	84.6	84.4	84.3	83.6	83.0	
Mean Daily Temperature Range	5% DB	MDBR	10.6	12.3	12.6	12.7	11.9	11.2	11.0	10.9	11.6	12.0	11.9	10.6	
		MCDBR	12.5	13.8	14.3	14.1	13.2	12.6	11.8	11.8	12.5	13.4	13.5	12.5	
		MCWBR	3.9	3.6	3.9	4.3	4.3	4.2	3.9	4.0	4.2	4.4	4.2	4.1	
	5% WB	MCDBR	11.0	12.3	12.7	12.8	12.0	11.3	11.3	11.1	11.7	12.0	12.1	11.3	
		MCWBR	4.1	3.8	4.4	4.6	4.5	4.5	4.3	4.4	4.5	4.6	4.3	4.1	
Clear-Sky Solar Irradiance	taub	0.470	0.483	0.493	0.492	0.478	0.490	0.482	0.524	0.555	0.581	0.485	0.470		
	taud	2.293	2.239	2.219	2.223	2.269	2.239	2.270	2.129	2.039	1.946	2.249	2.300		
	Ebn at Noon	273	271	267	261	258	251	255	250	247	242	267	271		
	Edh at Noon	43	47	48	46	43	43	42	50	56	62	45	43		
All-Sky Solar Radiation	RadAvg	1440	1664	1659	1609	1488	1412	1401	1422	1418	1421	1348	1314		
	RadStd	121	149	147	102	70	101	106	80	99	106	83	97		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (6 neighbors)	+0.67	+0.16	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page